Multitask Deep Learning for Segmentation and Lumbosacral Spine Inspection

Van Luan Tran, Huei-Yung Lin, *Senior Member, IEEE*, and Hsaio-Wei Liu

***Abstract*— Multitask learning (MTL) has achieved notable progress in many medical applications. In this article, we propose a multitask neural network, MRNet, for segmentation and spinal parameter inspection. It is developed based on the multipath convolutional neural network for the robust detection of obscure regions on X-ray images. The proposed MRNet has two branches. One is for the segmentation of lumbar vertebrae, sacrum, and femoral heads. It shares the main features with the second branch for detection and classification by supervised learning. The output of the second branch is used to estimate the parameters for lumbosacral spine inspection. We conduct this research on our dataset collected and annotated by doctors for model training and performance evaluation. The datasets are used to train our MRNet as well as other networks for performance evaluation. Compared to the state-of-the-art techniques, the proposed MRNet is capable of X-ray image segmentation and parameter estimation with very limited training data. The results have demonstrated the feasibility of our MRNet for the segmentation of lumbar vertebrae, as well as the automated parameter prediction for lumbosacral spine inspection. Code is available at https://github.com/LuanTran07/BiLUnet-Lumbar-Spine.**

***Index Terms*— Lumbosacral spine inspection, multitask learning (MTL), segmentation, X-ray image.**

I. Introduction

NOWADAYS, deep learning techniques have been extensively applied in many fields related to our daily lives, especially in medical applications [1]–[3]. One fundamental task is the segmentation and shape detection in medical images for object identification and location computation [4], [5]. While X-ray inspection is of great clinical necessity, the automatic understanding of X-ray images is still a very challenging task. It is primarily due to the projective nature of X-ray imaging, which causes large overlaps among the object shapes, blurred boundaries, and complicated composition patterns [6]. For specific medical applications such as the treatment for low back pain, the images of spine bones are acquired by X-ray [7], computed tomography [8], or magnetic resonance imaging [9], [10], to facilitate the assessment of the causes by doctors. Consequently, the automatic detection and segmentation of vertebrae in medical images is an important step for the analysis of the disease severity and recovery from the treatment.

The recent state-of-the-art techniques for vertebra detection, segmentation, and localization from medical images usually adopt two main strategies: 1) the approaches using conventional computer vision and image processing algorithms and 2) the methods based on deep learning paradigms. The former utilizes image enhancement, feature extraction, clustering, and classification for local region segmentation [11]–[15] and the latter adopts convolutional neural networks with the models trained using the datasets labeled by experts in the field [16]–[19]. Due to the overlapping shadows of chest organs, including lungs, bowels, and other bony structures such as ribs, automatic detection, and segmentation of vertebrae using traditional methods are more difficult for X-ray images compared to CT or MRI images [20], [21]. Nevertheless, X-ray imaging technologies are among the most frequently used clinical exams because of the least amount of radiation [22].

For vertebrae identification and localization, Chen *et al.* [23] proposed a framework to allow the fully convolutional networks (FCNs) to directly learn the long-range image patterns. This method enables the FCN classification to be trained at the vertebrae level and the localization issue to be decomposed into two different tasks that can be addressed at different scales. For the conventional approach, Zheng *et al.* [24] adopted Hough transform and Fourier descriptor to perform the vertebral extraction in digital videofluoroscopic (DVF) images. Huang *et al.* [25] used an adaptive boosting learning algorithm to extract vertebra regions in MRI images. The proposed method was able to achieve 98% and 96% on detection precision and segmentation accuracy for spine images, respectively. More recently, Korez *et al.* [26] presented a framework for automated detection and segmentation of the spine and vertebra on CT images. They applied an improved shape-constrained deformable model to provide high accuracy detection and segmentation. Nevertheless, its capability is limited by the variety of spinal structures, image modalities, and sensor's fields of view.

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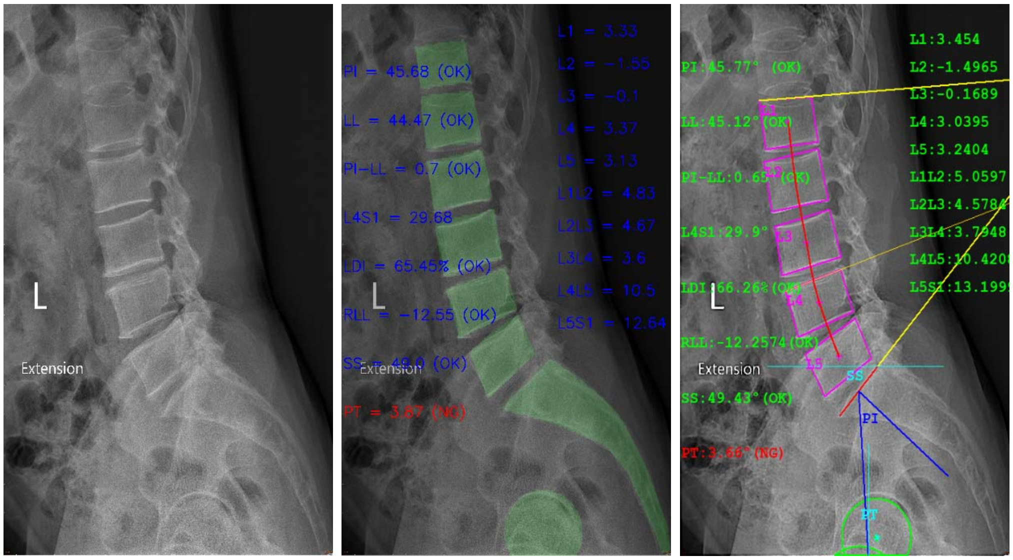


(a) (b) (c)

Fig. 1. Overview of the lumbosacral spine inspection on X-ray images by our multitask deep neural network for segmentation and inspection. (a) Input X-ray image. (b) Our MRNet generates the segmentation result and inspected values. The vertebrae, sacrum, and femoral heads are segmented with green color. (c) Annotation reference is provided by doctors.

Deep learning has achieved some significant advances in medical image analysis in recent years [27]–[30]. Specifically, multitask deep neural networks are used for various medical image segmentation tasks [31]–[34]. Multitask learning (MTL) performs model training for several tasks simultaneously to share information in the network and solve multiple tasks at the same time. Zhang *et al.* [27] proposed a multitask relational learning network for vertebral localization, identification, and segmentation on MRI images. This method combines two structures, the Co-Seg-Loc network, and a discriminator. The Co-Seg-Loc network serves the semantic localization regression, and the discriminator makes better predictions from the network and the ground truth by XOR loss. The proposed technique can boost the prediction accuracy on both the segmentation and localization tasks.

Recently, many researchers focus on dealing with the lumbar segmentation, and localization of sacrum, lumbar of L1–L5 [18], [22]. In BiLuNet [35], we trained on the image dataset for semantic segmentation of vertebrae, sacrum, and femoral heads. The post-processing was then performed to identify the shape and location of the targets (L1, L2, L3, L4, L5, S1, hip axis). The locations of five vertebrae (L1–L5), sacrum (S1), and hip axis (o) were used to derive the parameters for lumbosacral spine inspection. This method requires two steps to derive the parameters for lumbosacral spine inspection, and the post-processing is affected by the accuracy of the output. We aim to improve the accuracy by multitask deep learning to directly predict the inspected values for lumbosacral spine inspection. In MBNet [36], a multitask deep neural network based on the BiLuNet architecture for semantic segmentation is developed. It shares the main features with Inspected Values Branch to predict the inspected values for lumbosacral spine inspection. The results derived from MBNet are reported with higher accuracy.

In this article, we propose a multitask neural network for the binary segmentation of the lumbosacral spine as well as the parameter estimation for inspection and diagnosis. It is based on the effective segmentation network for the robust detection of unclear overlapping shadow regions on X-ray images. The parameters for the diagnosis of the lumbosacral spine are then derived. In the experiments, the training data are labeled at the pixel level manually and annotated by doctors. The datasets are used to train our MRNet, as well as other networks for performance comparison. Experimental results have demonstrated the effectiveness and feasibility of our proposed MRNet for both the segmentation and lumbosacral spine inspection on X-ray images with very high accuracy.

The highlight contributions of this work are as follows.

1) We propose a new architecture MRNet, which is a multitask deep neural network based on a convolutional neural network structure. The results are greatly improved, especially for the target obscured by other objects or not fully appeared in the image.
2) Compared to the state-of-the-art techniques, the proposed MRNet is capable of X-ray image segmentation with very limited training data.
3) Our method is able to achieve high accuracy for several tasks simultaneously by information sharing in the network with MTL. It is capable of performing the lumbosacral spine inspection on X-ray images.

II. Related Work

A. Multitask Deep Learning

In recent years, several multitask deep neural networks have been proposed for medical applications [37]–[39]. Liu *et al.* [40] proposed a multitask deep learning model for automated lung nodule analysis. They built a bi-branch

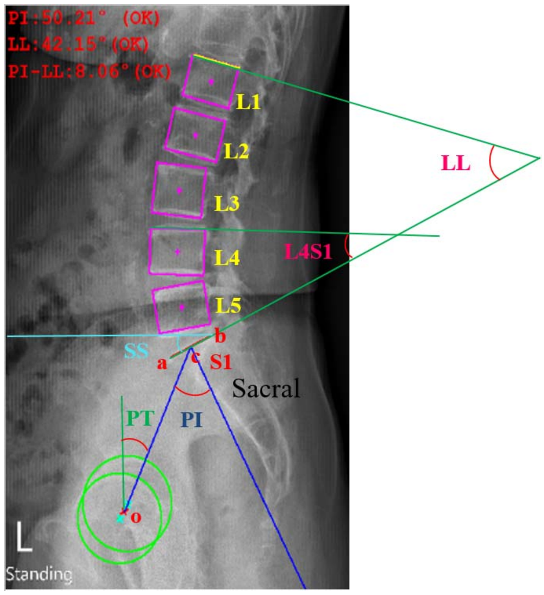


Fig. 2. Parameters used for lumbosacral spine inspection. S1 is defined by the line segment *ab* constructed between the posterior corner of the sacrum and the anterior tip of the S1 endplate. The hip axis (*o*) is the midpoint of the line segment connecting the centers of femoral heads. The pelvic radius is drawn from the hip axis to the center of S1.

model for classification and attributed score regression tasks. Their method makes the detection of marginal lung nodules more reliable for neural networks. Xiao *et al.* [41] proposed a new regularized MTL model to jointly select a small set of standard features across multiple tasks. The experiments demonstrated that the model provided better prediction results compared to other methods. Murugesan *et al.* [42] proposed a multitask network, Psi-Net, with a single encoder and three decoders to learn three different tasks in parallel. One decoder learns the calculation of segmentation masks, and the other two learn the auxiliary functions of contour identification and map estimation. Learning these adjunct tasks helps to catch the shape and the boundary detail, and consequently improves the segmentation performance.

B. Lumbosacral Spine

The lumbosacral spine inspection includes object segmentation, target localization, and analysis of several inclination angles in the X-ray images. Fig. 2 depicts the surgical measurements of interest in this work for identification and future examination. Pelvic incidence (PI) is defined as the angle between the line perpendicular to the sacral plate at its midpoint (c) and the line connecting this point to the axis of femoral heads [43]. PI is an important pelvic anatomic parameter that determines pelvic orientation. It is a specific constant for each individual and remains relatively stable during childhood. Afterward, it increases significantly during youth until reaching its maximum, and does not change after adolescence [44].

Sacral slope (SS) is defined as the angle between the sacral endplate S1 and the horizontal axis passing through its endpoint (b). Pelvic tilt (PT) is defined as the angle between the line passing through the hip axis o and the center of S1, and a vertical line in the standing position (see Fig. 2). PT and SS are position-dependent angles in the spatial orientation of the pelvis [45]. PI, SS, and PT are particularly useful since it can be shown that PI is the arithmetic sum of SS and PT, and these two position-dependent variables can determine the pelvic orientation in the sagittal plane [46].

Lumbar lordosis (LL) is the inward curve of the lumbar spine and is defined as the angle between the lines drawn along the superior endplate of L1 and the superior endplate of S1 [47]. The angle L4S1 is measured by the intersection of the lines drawn along the superior endplate of L4 and the superior endplate of S1. Most bodyweight and movement are carried by the two lowest segments in the lumbar spine, L5/S1, and L4/L5, which makes the area prone to injury. In 95% of disk herniated cases, L4/L5 or L5/S1 levels are involved. Low back pain with degenerative disk disease is when normal changes take place in the disks of the spine, mostly at the sites L4/L5 lumbar vertebrae and S1 sacrum [48], [49]. The surgery for patients with unbalanced lumbar spondylolisthesis is helpful to correct the spine-pelvic sagittal parameters (PI, PT, SS, L4S1, and LL) closer to the normal range [50].

PI-LL is the value of PI minus LL. Given a PI value, it offers an estimate of the lordosis and quantifies the mismatch between pelvic morphology and the lumbar curve. Lordosis distribution index (LDI) defines the magnitude of the arc lordosis (L4S1) in proportion to the arc of the lumbar lordosis (LL). Relative lumbar lordosis (RLL) quantifies the magnitude of lumbar lordosis relative to the ideal lordosis as defined by the magnitude of PI. RLL and LDI are used to describe the state where the lumbar lordosis is rehabilitated to the ideal magnitude and distribution. The evaluation of RLL and PI-LL provides the prediction of postoperative complications and the correlation with health-related quality of life (HRQOL) scores [51].

The measurement of vertebral bodies reveals kyphotic values for L1, neutral values for L2, and gradually lordotic values for L3 to L5 [52]. As shown in Fig. 2, the intervertebral disk curvature angle (L1L2) is the angle between the L1 lower edge and L2 upper edge. They are calculated similarly for L2L3, L3L4, and L4L5. The wedge angle (L5S1) is the angle between L5 lower edge and S1 (the superior endplate). From L1L2, the intervertebral disk shows a gradual lordotic angulation. Nearly 60% of the angular measurement of lumbosacral curvature is accounted by caudal parts of the curvature, intervertebral disks L4L5 and L5S1, and the vertebral body L5. For the vertebral bodies L2 and L4 and the lumbar curvature measurements, a substantial difference is between males and females. In the head-tail direction of lumbosacral curvatures, vertebral bodies and intervertebral disks present a steady increase in lordotic participation. The vertebral body L1 is the only lumbar curvature feature with medium kyphotic participation. The percent participation ranges from L1 to L4 for vertebral bodies, as well as L1L2 for intervertebral disks. The individuals with kyphotic bends in the vertebral bodies and intervertebral disks are found to be responsible for this observation. In all patients, only the vertebral body L5 and the intervertebral disks L2L3 through L5S1 demonstrate the lordotic bend.

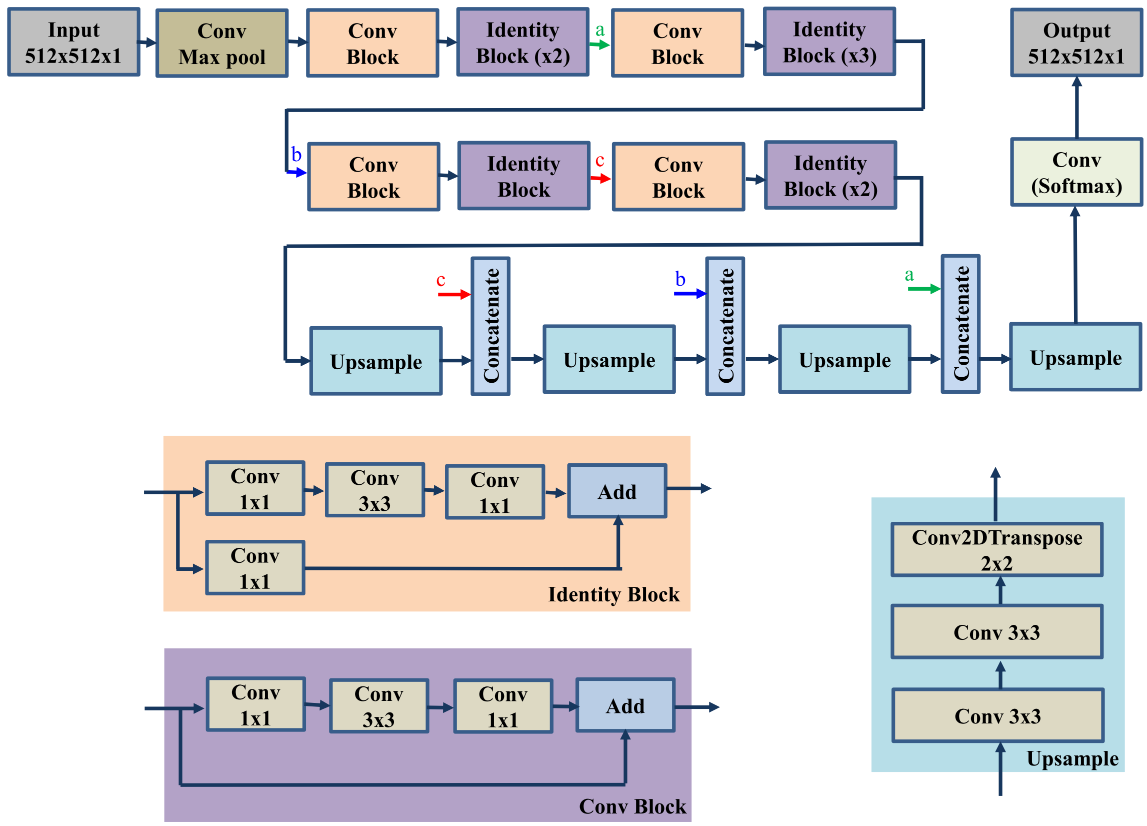


Fig. 3. End-to-end architecture of the segmentation branch. This architecture simulates our network model with three modules of identity block, Conv block, and upsample block.

III. Multitask Deep Neural Network

In this section, we present the proposed multitask network architecture. It automatically learns appropriate representation sharing through end-to-end training, and makes full use of the correlations among different tasks. As illustrated in Fig. 5, our network model constructed with supervised learning is applied for two tasks, segmentation and parameter inspection for the diagnosis of lumbar vertebrae. In the proposed MRNet architecture, M stands for the multitask network structure and R denotes the model developed based on ResNet [53].

A. Deep Neural Network

For image segmentation, this work is specifically concerned with the method to extract accurate object masks. It is used to identify the locations of lumbar vertebrae, sacrum, and femoral heads. In this regard, ResNet is one of the most popular FCNs available in the existing research. It is also the earliest architecture that adopts batch normalization. The previous network models often improve the accuracy by increasing the depth of convolutional neural networks. When traversing too many layers of depth, it can cause the original information to be lost. Some researchers deal with this problem by using shortcut connections to keep the information from the earlier layers to the later layers [54]. The convolution block consists of two convolution branches where one branch applies a 1×1 convolution before being added directly to the other branch. The identity block does not apply the 1×1 convolution, but directly adds the value of that branch to the other branch.

Fig. 3 depicts the segmentation branch of our network used to extract the features and share for the inspected value branch. In this segmentation branch, a deep convolutional network is built based on [53]. It contains the input layer of $512 \times 512 \times 1$, and five stages for downsampling the features. The first stage consists of the convolution of 64 output filters with the shape of $(3, 3)$, stride $(2, 2)$, BatchNorm, and MaxPooling $(2, 2)$. The second stage has a convolutional block using three filters with the sizes of $(64, 64, 256)$. This stage takes two identity blocks with the filters of $(64, 64, 256)$. The third stage consists of a convolutional block using three filters with the size of $(128, 128, 512)$. This stage has three identity blocks with filters of $(128, 128, 512)$. The fourth stage contains a convolutional block using three filters with the size of $(256, 256, 1024)$. This stage has one identity block with filters of $(256, 256, 1024)$. In the final stage, a convolutional block using three filters with the size of $(512, 512, 2048)$ is adopted. This stage consists of two identity blocks with filters of $(512, 512, 2048)$.

The output of the final stage is connected to an Upsample module. It has two 3×3 convolutions with 512 output filters. A conv2Dtranspose is then applied with 256 output filters using the 2×2 kernel and stride $(2, 2)$. The output of the Upsample module is concatenated with the output of the identity block in the fourth stage. It is then upsampled to concatenate with the output of the identity block in the

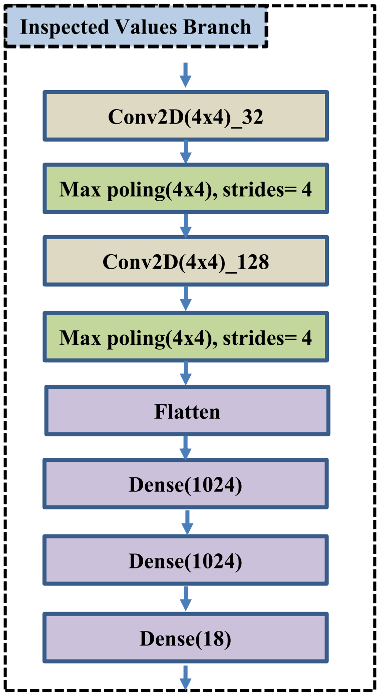


Fig. 4. Inspected values branch merged into FC layers to learn the parameter prediction for inspection and diagnosis.

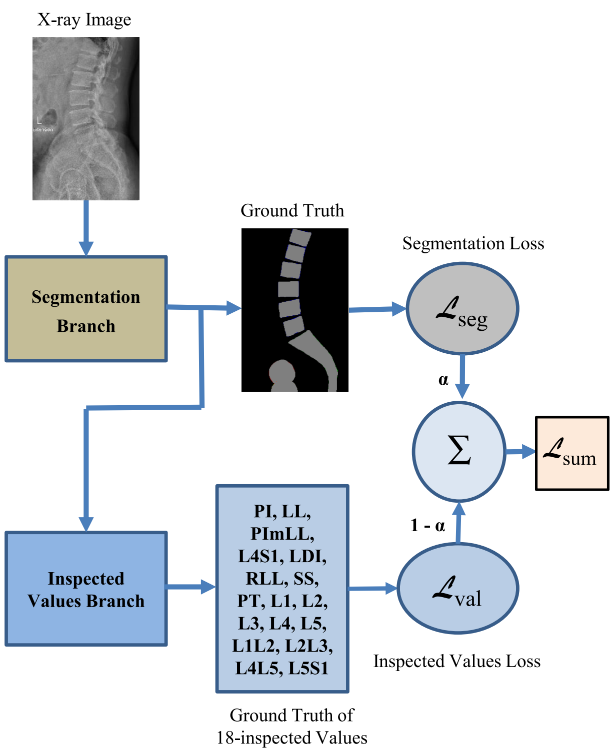


Fig. 5. MRNet architecture consists of two branches. The segmentation branch is trained to produce a segmentation of the lumbar vertebrae, sacrum, and femoral heads. The inspected values branch generates the inspected values for the related parameters for the diagnosis of the lumbar vertebrae.

third stage. Finally, the next output is upsampled again to concatenate with the output of the identity block in the second stage. The output of the Upsample module is connected to a convolutional layer with the activation using softmax. In the final layer, the output has the same size as the original image of 512 × 512 × 1.

TABLE I

PERFORMANCE COMPARISON OF MIOU ON OUR DATASET FOR DIFFERENT ALGORITHMS, PSPNET [55], U-NET [56], UNET++ [57], BILUNET [35], AND OUR MODEL ON THE TESTING DATASET

Methods	mIoU
PSPNet	79.3%
U-Net	80.1%
UNet++	84.7%
BiLuNet	86.1%
Ours	88.5%

TABLE II

STANDARD RANGE FOR OUR LUMBOSACRAL SPINE INSPECTION

Parameters	Standard range
PI	34°∼84°
LL	31°∼79°
PImLL	-10°∼10°
LDI	50%∼80%
RLL	-14°∼11°
SS	-20°∼65°
PT	5°∼30°

In addition to generating the segmentation map, our network is also constructed to derive the physical quantities. It is used to estimate the parameters automatically for doctors to perform clinical diagnoses of low back pains. The classification task is considered an auxiliary task here. While the segmentation task provides local supervision for lumbar vertebrae, sacrum, and femoral heads from clinical X-ray images, the regression values for the level classification task provide global supervision about how inspected values. The integrated classification approach provides the model more robust to the variation of inspected values and increases the performance. Via the use of an image dataset, segmentation mask, and inspected value ground truth, we train our network to predict the parameters used for lumbosacral spine inspection.

As shown in Fig. 5, the input of the inspected values branch is shared from the final layer of the segmentation branch. The inspected values branch is designed to extract from the shared feature maps. This branch consists of two 4 × 4 convolutional layers, two 4 × 4 max-pooling layers, and three fully connected (FC) layers constructed as an inspected value classifier. The first and second FC layers have 1024 neurons with a sigmoid activation function. The output FC layer contains 18 neurons and is followed by a sigmoid activation function, which indicates the probabilities of the 18-inspected values as shown in Fig. 4. For the supervised training of inspected value classifier, the mean squared error (MSE) regression loss is given by (3) is used.

B. Multiple Loss Function

Our loss function is a weighted sum of a segmentation loss (L_{seg}) and an inspected values loss (L_{val})

$$L_{sum} = \alpha \cdot L_{seg} + (1 - \alpha) \cdot L_{val} \quad (1)$$

where $\alpha \in [0, 1]$ is a loss weight. It adjusts the relative importance of L_{seg} and L_{val}. The loss weight α is set to 0.9 in our method.

We use the binary cross-entropy loss function for segmentation loss. This function is used to solve binary classification

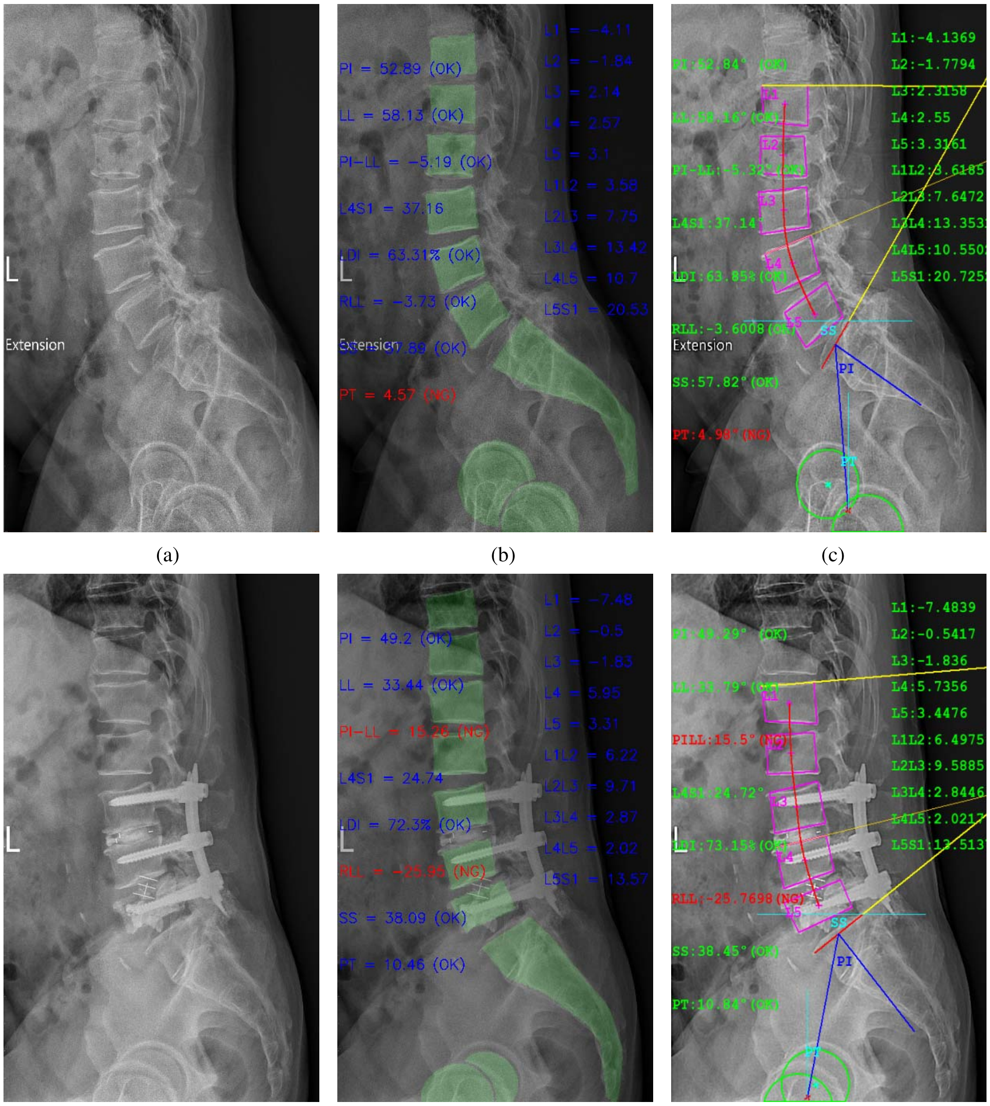


Fig. 6. Testing results of lumbosacral spine inspection on X-ray images. (a) and (d) Input X-ray images, (b) and (e) parameter inspection results and the segmentation of vertebrae, sacrum, left and right hip joints by our MRNet (marked in green), and (c) and (f) reference annotation provided by a doctor.

problems with only two classes and is given by

$$L_{\text{seg}} = -\frac{1}{N}\sum_{i=1}^{N}(y_i \cdot \log(\hat{y}_i) + (1-y_i)\cdot\log(1-\hat{y}_i) \quad (2)$$

where y_i and $\hat{y}_i$ denote the actual labels of the data point and the predicted probability of the data point in the image map, respectively. N is the total number of data points in the image map.

The inspected value loss is defined by MSE regression loss. It is the most commonly used regression loss function. The MSE is the sum of squared distances between our target variable and predicted values. The inspected values loss is given by

$$L_{\text{val}} = \frac{\sum_{i=1}^{n}(y_j - \hat{y}_j)^2}{n} \quad (3)$$

where y_j and $\hat{y}_j$ denote the predicted and ground-truth values of the inspected values in the X-ray image, respectively. n is the total number of training samples.

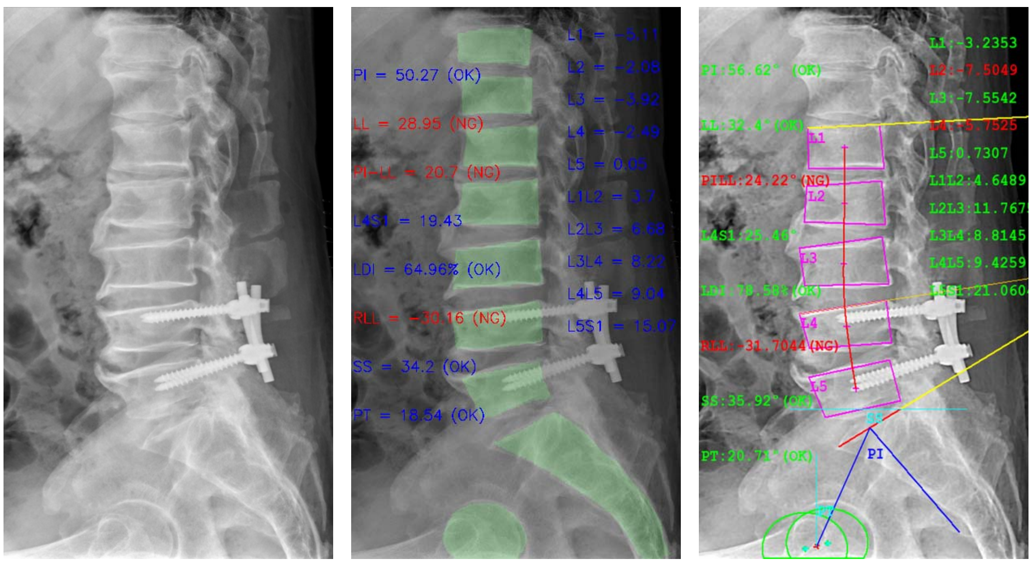
IV. Experiments

The dataset in our experiments consists of 900 images with half pre-surgery and half post-surgery for network training. An additional 100 images are added for testing, with also half pre-surgery and half post-surgery. The training images are labeled with the regions of vertebra, sacrum, femoral_head1, femoral_head2, and ground-truth, and 18 inspected values are annotated by doctors. The computation is carried out on a computer with an Intel i7-7700 CPU at 3.6 GHz and 32-GB RAM. The GPU is an NVIDIA GeForce RTX 3060 with 12 GB RAM. For the network training, the number of epochs and the batch size are set as 200 and 4, respectively. The initial learning rate is set as 3e-4 for Adam.

The loss weight α is used to adjust the relative importance of the segmentation loss and the inspected value loss. In our method, the segmentation shares feature for the inspected values branch. It also has effects on the inspected value accuracy. For the total loss with $\alpha = 0.8$, the segmentation accuracy is low, and so is the inspected value

TABLE III
PERFORMANCE EVALUATION ON THE TESTING IMAGE SHOWN IN FIG. 6(b) (PRESURGERY) FOR OUR MRNET, MBNET [36], AND BILUNET [35]

Names	MBNet	BiLuNet	Ours	GT	Acc MBNet (%)	Acc BiLuNet (%)	Acc Ours (%)
PI°	52.48	54.40	52.89	52.84	99.28	96.88	99.90
LL°	58.28	54.55	58.13	58.16	99.75	92.48	99.94
PImLL°	-3.97	-0.15	-5.19	-5.32	93.25	74.15	99.35
LDI%	65.30	68.30	63.31	63.85	95.17	85.17	98.20
RLL°	-4.58	-8.18	-3.73	-3.60	96.08	81.68	99.48
SS°	57.30	57.55	57.89	57.82	99.39	99.68	99.92
PT°	4.67	-3.15	4.57	4.98	98.76	67.48	98.83
L1°	-3.73	-4.07	-4.11	-4.14	90.1	98.31	99.28
L2°	-1.78	-3.72	-1.84	-1.78	83.15	-	96.63
L3°	2.25	-7.31	2.14	2.30	97.83	-	93.04
L4°	2.63	1.85	2.57	2.55	96.86	72.55	99.22
L5°	3.67	8.82	3.10	3.32	89.46	-	93.37
L1L2°	3.57	1.27	3.58	3.62	98.62	35.08	98.90
L2L3°	6.92	14.86	7.75	7.65	90.46	5.75	98.69
L3L4°	13.65	16.26	13.42	13.35	97.75	78.20	99.48
L4L5°	10.91	9.74	10.70	10.55	96.59	92.32	98.58
L5S1°	20.36	16.84	20.53	20.73	98.22	81.23	99.04



(a) (b) (c)

Fig. 7. Testing results of a challenging case where the second femoral head is hardly observed in the image. (a) Input X-ray images, (b) parameter inspection results and the segmentation of vertebrae, sacrum, left and right hip joints by our MRNet (marked in green), and (c) reference annotation provided by a doctor.

accuracy. Thus, we set the loss weight $\alpha = 0.9$ empirically. Our experimental results show that the joint training of two networks has a better performance than training sequentially.

The segmentation is carried out for object regions of vertebra, sacrum, femoral_head1 and femoral_head2, and represented in the green color (see Figs. 6 and 7). As shown in Fig. 6(b) and (e), the inspection results of our model are very close to the annotation provided by doctors [Figs. 6(c) and (c)]. In Table I, the performance comparison in terms of mIoU is presented with different algorithms. The testing data for evaluation contain 100 images with a half pre-surgery and half post-surgery. The mIoU derived from our network model is 88.5, which is higher than the previous methods including PSPNet, U-Net, UNet++, and BiLuNet.

Table II shows the standard ranges of common interests, PI, LL, PImLL, LDI, RLL, SS, and PT [58]. The angles PI and PT are determined by the center of S1 and the hip axis, so they are affected by the locations of femoral heads and S1. More specifically, the derivation of most parameters depends on the accuracy of S1. PImLL (PI – LL) presents an appreciation of the lordosis required for a given PI value and quantifies the mismatch between pelvic morphology and the lumbar curve [51]. The value LDI and the angle RLL are defined by

$$\mathrm{LDI} = (\mathrm{L4S1}/\mathrm{LL}) \tag{4}$$

$$\mathrm{RLL} = \mathrm{LL} - (0.62 \cdot \mathrm{PI} + 29). \tag{5}$$

In the experiments, the inspection results are marked as OK if the values are in the standard range.

To evaluate the lumbosacral spine inspection with respect to the annotated values (ground-truth), the accuracy is defined as

$$\mathrm{Acc} = \left(1 - \frac{|v_e - v_d|}{v_s}\right) \cdot 100\% \tag{6}$$

TABLE IV

PERFORMANCE EVALUATION ON THE CHALLENGING IMAGE (POSTSURGERY) WITH THE SECOND FEMORAL HEAD SHOWN IN FIG. 7 FOR OUR MRNET, MBNET [36], AND BILUNET [35]

Names	MBNet	BiLuNet	Ours	GT	Acc MBNet(%)	Acc BiLuNet (%)	Acc Ours (%)
PI°	44.99	38.29	50.27	56.62	98.44	85.04	87.30
LL°	45.03	39.74	28.29	32.40	99.81	88.79	91.40
PImLL°	0.46	-1.45	20.70	24.22	99.05	89.50	92.67
LDI%	65.48	62.23	64.96	78.58	97.40	86.57	82.40
RLL°	-13.2	-13.00	-30.16	-31.7	96.24	97.04	93.84
SS°	49.03	46.04	34.20	35.92	99.53	96.01	97.98
PT°	4.09	-7.75	18.54	20.71	98.28	54.36	93.80
L1°	-5.26	2.37	-5.11	-3.24	37.65	73.15	42.28
L2°	-3.12	0.58	-2.08	-7.51	41.54	7.72	27.70
L3°	-3.87	9.52	-3.92	-7.55	51.26	73.91	51.92
L4°	-2.12	7.41	-2.49	-5.75	36.87	71.13	43.30
L5°	0.10	13.36	0.05	0.73	13.70	-	6.85
L1L2°	3.60	0.16	3.70	4.65	77.42	3.44	79.57
L2L3°	5.98	0.74	6.68	11.8	50.68	6.27	56.61
L3L4°	6.22	-0.77	8.22	8.81	70.60	8.74	93.30
L4L5°	8.17	-7.09	9.04	9.43	86.64	75.19	95.86
L5S1°	14.68	18.7	15.07	15.07	69.71	88.79	71.56

TABLE V

PERFORMANCE EVALUATION ON THE DATASET WITH 100 PRESURGERY AND POSTSURGERY X-RAY IMAGES FOR OUR MRNET, MBNET [36], AND BILUNET [35]

Names	Acc MBNet(%)	Acc BiLuNet (%)	Acc Ours (%)
PI°	94.48	88.30	95.60
LL°	92.10	89.24	91.81
PImLL°	81.07	78.14	87.23
LDI%	78.55	83.90	82.24
RLL°	86.63	81.36	89.30
SS°	91.29	90.10	92.06
PT°	91.15	83.77	93.16
L1°	80.96	65.24	81.43
L2°	83.89	54.39	84.08
L3°	88.05	70.30	87.28
L4°	90.10	72.09	90.30
L5°	73.26	51.12	75.68
L1L2°	86.19	69.13	89.57
L2L3°	85.30	85.06	87.61
L3L4°	89.87	81.84	91.78
L4L5°	91.03	79.68	92.13
L5S1°	84.97	81.01	86.24

where υ_e and υ_d are the value derived from our lumbosacral spine inspection and the value annotated by doctors, respectively, and υ_s is the standard range shown in Table II.

The evaluation results of lumbosacral spine inspection for specific cases are shown in Tables III and IV. These tables report the accuracy of the images shown in Figs. 6 and 7. In these tables, the accuracy of L1, L2, L3, L4, L5, L1L2, L2L3, L3L4, L4L5, and L5S1 are evaluated by

$$\mathrm{Acc} = \left(1 - \frac{|\upsilon_e - \upsilon_d|}{\upsilon_d}\right) \cdot 100\% \tag{7}$$

where υ_e and υ_d are the value from our inspection result and the value annotated by doctors, respectively.

In BiLuNet [35], we trained on this X-ray image dataset for semantic segmentation of vertebrae, sacrum, and femoral heads. Moreover, the segmentation of femoral heads is a challenging task due to the overlap of pelvic and femoral heads. The femoral heads are used to determine the hip axis o and other parameters, and the inspected values of PI, PT, PImLL, and RLL are affected accordingly. The experimental results are reported in Table V. In MBNet [36], we also used this dataset for training. It is a multitask deep neural network based on the BiLuNet architecture for semantic segmentation. The results derived from MBNet are with higher accuracy as compared in Table V.

In this work, the proposed MRNet is trained and tested with binary segmentation X-ray images. The network model output is for the segmentation of vertebrae, sacrum, and femoral heads. The main features are then shared with the Inspected Values Branch to predict the inspected values for lumbosacral spine inspection. Table V tabulates the evaluation results of MRNet, MBNet, and BiLuNet tested on 100 X-ray images. It indicates that MRNet achieves a high average accuracy on PI, LL, PImLL, LDI, RLL, SS, and PT with 95.60%, 91.81%, 87.23%, 82.24%, 89.30%, 92.06%, and 93.16%, respectively. The average accuracy of vertebral body angles for L1, L2, L3, L4, and L5 are 81.43%, 84.08%, 87.28%, 90.30%, and 75.68%, respectively. Finally, the average accuracy of intervertebral disk curvature angles for L1L2, L2L3, L3L4, L4L5, and the wedge angle (L5S1) are 89.57%, 87.61%, 91.78%, 92.13%, and 86.24%, respectively.

In addition to the performance improvement of the proposed MRNet compared to the previous approaches, the evaluation results reported in Table V also indicate that the lumbosacral spine inspection is highly affected by the segmentation accuracy. BiLuNet has the parameter inspection of PT, PI, PImLL, and RLL affected by the femoral head locations and presents less accuracy. In general, the X-ray images with femoral heads unclear or missing are considered some challenging cases for parameter inspection. One typical example is shown in Fig. 7, where the second femoral head is not available in the image for segmentation. In such cases, MRNet is still able to provide accurate parameter estimates.

V. CONCLUSION

In this article, we proposed a multitask deep neural network for the segmentation of lumbar vertebrae, sacrum, and femoral heads on pre-surgery and post-surgery X-ray images, as well as the parameter prediction for inspection and diagnosis. A new convolutional network structure is presented for the segmentation of X-ray images. It can achieve high accuracy for automated segmentation of lumbar vertebrae, sacrum, and femoral heads. With the MRNet architecture for MTL, accurate parameters are derived for the clinical diagnosis of low back pains. The experiments have demonstrated that our technique is able to achieve high accuracy on lumbosacral spine inspection. The proposed MRNet provides the comparative accuracy levels for both the pre-surgery and post-surgery X-ray images.

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